

HYPERTEXT MARKUP LANGUAGE

HTML Forms

HTML forms are interactive sections of a webpage that allow users to enter, select, or submit information. Forms act as a bridge between visitors and web applications by collecting data and sending it for processing.

Whether a user logs in, registers, searches, places an order, or submits feedback, a form is usually involved behind the scenes.

Why Forms Are Important

Forms make webpages interactive rather than purely informational.

They allow users to:

- Create accounts
- Sign in to websites
- Submit contact information
- Search for content
- Upload files
- Place online orders
- Complete surveys
- Send feedback

Without forms, websites would only display information and could not receive input from users.

Forms Introduction

An HTML form is created using the `<form>` element.

The form acts as a container that holds different input controls such as text boxes, buttons, checkboxes, and dropdown menus.

Basic Syntax

`<form>`

Form Elements Here

`</form>`

Structure of a Form



Simple Form Example

```
<form>
  <label>Name:</label>
  <input type="text">
  <br><br>
  <input type="submit">
</form>
```

This form asks for a user's name and provides a submit button.



The image shows a side-by-side comparison. On the left is a code editor window titled 'hypertext markup language.html'. It displays the following HTML code:

```
1 <!DOCTYPE html>
2 <html>
3 <head>
4   <title>FORM EXAMPLE</title>
5 </head>
6 <body>
7   <form>
8     <label>Name:</label>
9     <input type="text">
10    <br><br>
11    <input type="submit">
12  </form>
13 </body>
14 </html>
```

On the right is a web browser window titled 'FORM EXAMPLE'. The address bar shows 'File D:/hypertext%20markup%20language.html'. The page content displays a form with the label 'Name:' followed by a text input field, two blank lines, and a 'Submit' button.

This form asks for a user's name and provides a submit button.

Input Types:

The `<input>` element is the most commonly used form control. Different input types collect different kinds of information.

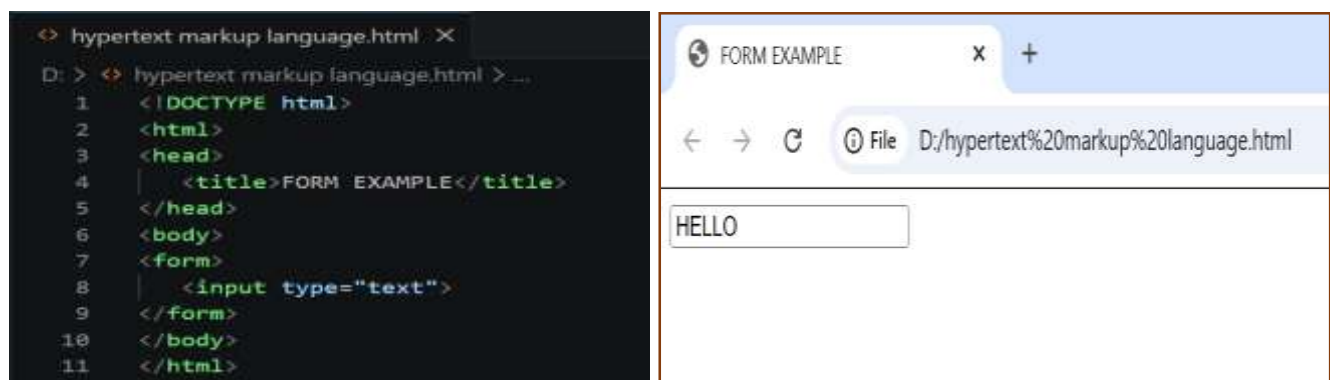
1. Text Input

Used for short text.

```
<input type="text">
```

Examples:

- Name
- City
- Username



The image shows a side-by-side comparison. On the left is a code editor window titled 'hypertext markup language.html'. It displays the following HTML code:

```
1 <!DOCTYPE html>
2 <html>
3 <head>
4   <title>FORM EXAMPLE</title>
5 </head>
6 <body>
7   <form>
8     <input type="text">
9   </form>
10 </body>
11 </html>
```

On the right is a web browser window titled 'FORM EXAMPLE'. The address bar shows 'File D:/hypertext%20markup%20language.html'. The page content displays a form with a single text input field containing the text 'HELLO'.



2. Password Input

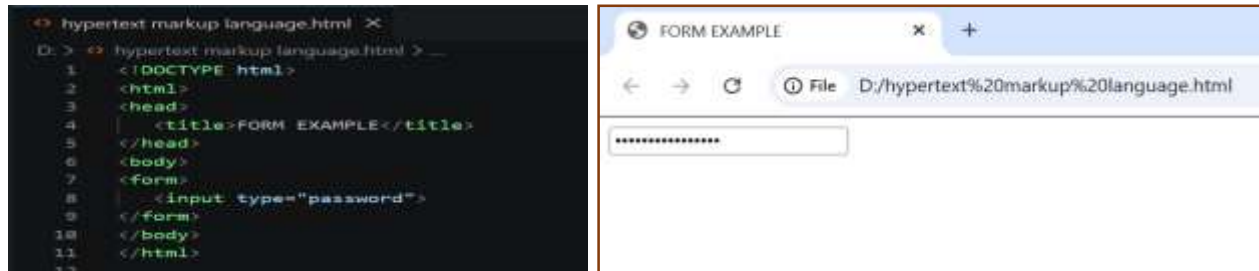
Hides entered characters.

`<input type="password">`

Example:

Useful for:

- Passwords
- PINs



3. Email Input

Accepts email addresses.

`<input type="email">`

Example:

user@example.com



Browsers can automatically verify the format.

4. Number Input

Accepts numeric values only.

`<input type="number">`

Examples:

- Age
- Quantity
- Marks



5. Date Input

Allows date selection.

`<input type="date">`

Example:

2026-05-30



6. Time Input

Allows time selection.

`<input type="time">`

Example:

10:45 AM



7. Radio Button

Used when only one option can be selected.

`<input type="radio" name="gender">`

Example:

☐ Male

☐ Female

☐ Other

Only one choice is possible.



8. Checkbox

Allows multiple selections.

`<input type="checkbox">`

Example:

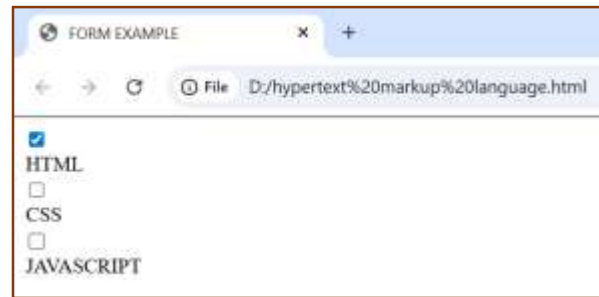
✓ HTML

✓ CSS

✓ JavaScript

Multiple options may be selected.

```
hypertext markup language.html X
D: > hypertext markup language.html > ...
1 <!DOCTYPE html>
2 <html>
3 <head>
4 | <title>FORM EXAMPLE</title>
5 </head>
6 <body>
7 <form>
8 | <input type="checkbox">
9 | <option>HTML</option>
10 | <input type="checkbox">
11 | <option>CSS</option>
12 | <input type="checkbox">
13 | <option>JAVASCRIPT</option>
14 </form>
15 </body>
16 </html>
```



9. File Upload

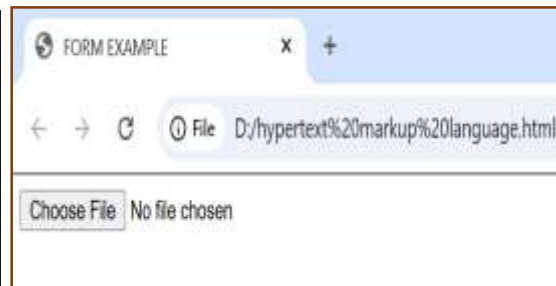
Allows users to upload files.

`<input type="file">`

Used for:

- Images
- Documents
- Videos

```
hypertext markup language.html X
D: > hypertext markup language.html > ...
1 <!DOCTYPE html>
2 <html>
3 <head>
4 | <title>FORM EXAMPLE</title>
5 </head>
6 <body>
7 <form>
8 | <input type="file">
9 </form>
10 </body>
11 </html>
```



10. Color Picker

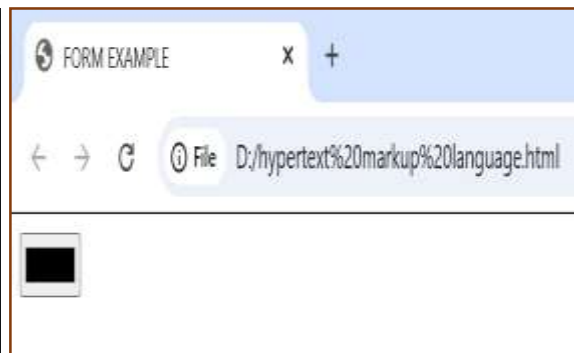
Allows color selection.

`<input type="color">`

Example:

Choose Favorite Color

```
hypertext markup language.html X
D: > hypertext markup language.html > ...
1 <!DOCTYPE html>
2 <html>
3 <head>
4 | <title>FORM EXAMPLE</title>
5 </head>
6 <body>
7 <form>
8 | <input type="color">
9 </form>
10 </body>
11 </html>
```



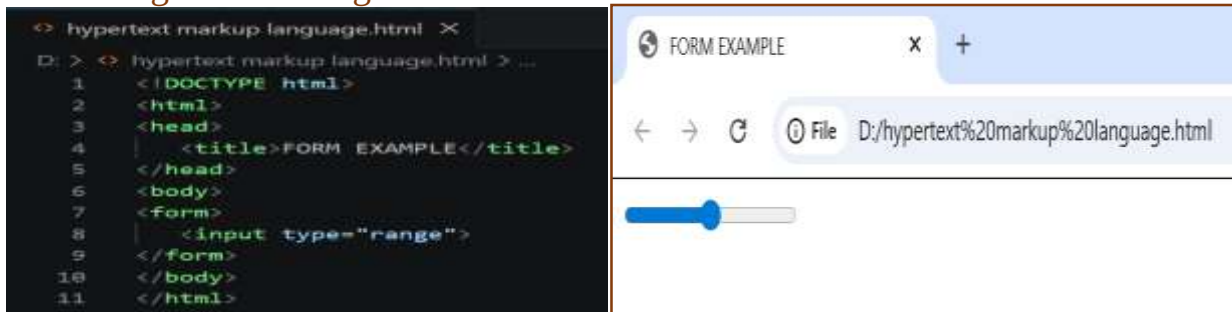
11. Range Slider

Creates a draggable slider.

`<input type="range">`

Useful for:

- Ratings
- Volume controls
- Brightness settings



12. Search Box

Designed for search operations.

`<input type="search">`

Example:

Search Products...



13. URL Input

Accepts website addresses.

`<input type="url">`

Example:

<https://example.com>

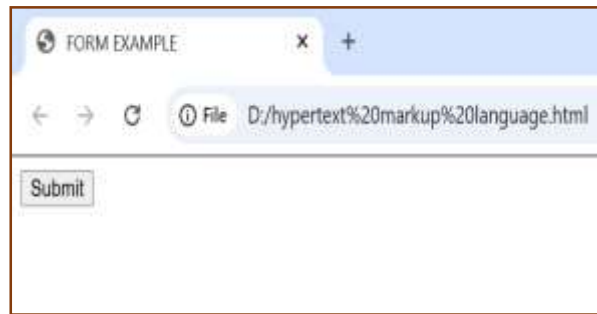


14. Submit Button

Sends form data.

`<input type="submit">`

```
hypertext markup language.html X
D: > <> hypertext markup language.html > ...
1  <!DOCTYPE html>
2  <html>
3  <head>
4    <title>FORM EXAMPLE</title>
5  </head>
6  <body>
7    <form>
8      <input type="submit">
9    </form>
10 </body>
11 </html>
```

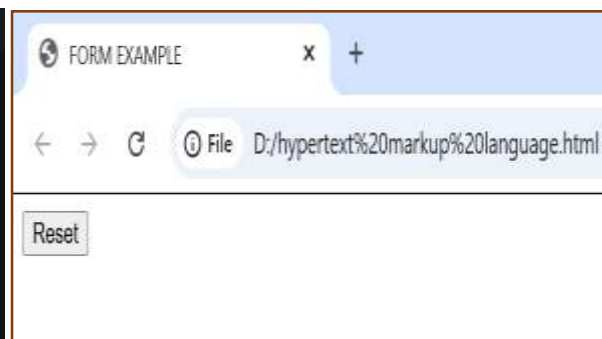


15. Reset Button

Clears all entered data.

`<input type="reset">`

```
hypertext markup language.html X
D: > <> hypertext markup language.html > ...
1  <!DOCTYPE html>
2  <html>
3  <head>
4    <title>FORM EXAMPLE</title>
5  </head>
6  <body>
7    <form>
8      <input type="reset">
9    </form>
10 </body>
11 </html>
```



Other Form Elements

Forms are not limited to `<input>`.

HTML provides additional controls.

Textarea

Used for longer text.

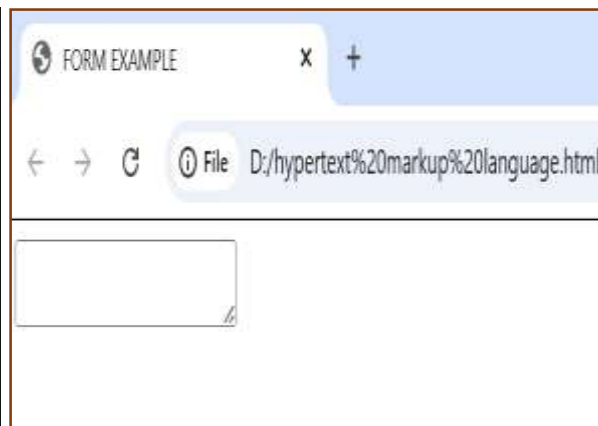
`<textarea>`

`</textarea>`

Examples:

- Feedback
- Comments
- Messages

```
hypertext markup language.html X
D: > <> hypertext markup language.html > ...
1  <!DOCTYPE html>
2  <html>
3  <head>
4    <title>FORM EXAMPLE</title>
5  </head>
6  <body>
7    <form>
8      <textarea>
9    </textarea>
10 </form>
11 </body>
12 </html>
```



Select Dropdown

Provides predefined options.

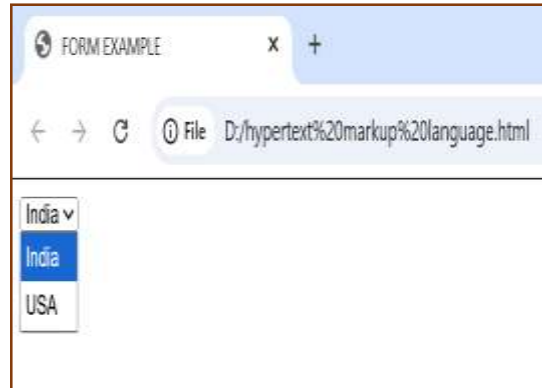
`<select>`

`<option>India</option>`

`<option>USA</option>`

`</select>`

```
hypertext markup language.html X
D: > <> hypertext markup language.html > ...
1  <!DOCTYPE html>
2  <html>
3  <head>
4    <title>FORM EXAMPLE</title>
5  </head>
6  <body>
7    <form>
8      <select>
9        <option>India</option>
10       <option>USA</option>
11      </select>
12    </form>
13  </body>
14 </html>
15
```



Button Element

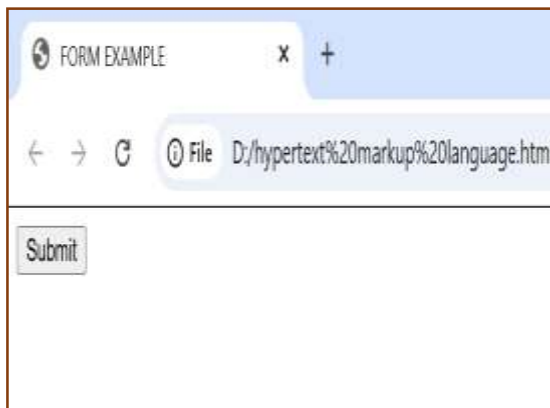
Creates a clickable button.

`<button>`

Submit

`</button>`

```
hypertext markup language.html X
D: > <> hypertext markup language.html > ...
1  <!DOCTYPE html>
2  <html>
3  <head>
4    <title>FORM EXAMPLE</title>
5  </head>
6  <body>
7    <form>
8      <button>
9        Submit
10     </button>
11  </form>
12 </body>
13 </html>
```



Form Validation

Validation checks whether entered information is acceptable before submission. It helps prevent incomplete or incorrect data.

Why Validation Matters

Validation ensures:

- Required fields are filled
- Correct formats are used
- Invalid data is prevented
- User errors are reduced

Required Field Validation

```
<input type="text" required>
```

The form cannot be submitted if the field is empty.

Email Validation

```
<input type="email">
```

Browser checks email format automatically.

Valid:

user@gmail.com

Invalid:

usergmail.com

Minimum and Maximum Values

```
<input type="number"
```

```
  min="1"
```

```
  max="100">
```

Only values between 1 and 100 are accepted.

Length Validation

```
<input
```

```
type="text"
```

```
minlength="3"
```

```
maxlength="15">
```

Limits text length.

Pattern Validation

Uses regular expressions.

```
<input
```

```
type="text"
```

```
pattern="[A-Za-z]+">
```

Accepts only alphabetic characters.

Placeholder Text

Provides input hints.

```
<input
```

```
type="text"
```

```
placeholder="Enter Name">
```

Displayed until the user begins typing.



Labels and Accessibility

Labels describe the purpose of form fields.

Accessibility ensures everyone can use forms effectively, including people using screen readers or keyboards.

The Label Element

```
<label>Name</label>
```

```
<input type="text">
```

This identifies the field's purpose.

Associating Labels with Inputs

```
<label for="name">
```

Full Name

```
</label>
```

```
<input
```

```
type="text"
```

```
id="name">
```

Benefits:

- Better usability
- Better accessibility
- Easier field selection

Why Labels Matter

Without Labels:

[_____]

User may not know what to enter.

With Labels:

Name:

[_____]

Accessibility Features

Good forms should:

- Use labels for every field
- Use meaningful button text
- Provide error messages
- Support keyboard navigation
- Maintain clear structure

